

ERNESTO BONILLA-HOSHIKAWA

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OBJECTIVE

Seeking a Fall 2026 co-op in embedded systems, robotics, AI, or software engineering, applying experience with FPGA development, C++, hardware-software integration, and backend systems in hands-on engineering environments.

EDUCATION

Northeastern University, Boston, MA

B.S. in Computer Engineering & Computer Science

Expected May 2028

Honors: GPA 3.97; Merit Scholarship Recipient; Tau Beta Pi Engineering Honor Society.

Relevant Coursework: Embedded Systems, Circuits & Signals, OOP, Fundamentals of Networks, Logic & Computation, Discrete Math, Differential Equations, Physics: Mechanics & EM.

Saint Peter's Preparatory School, Jersey City, NJ

September 2020 – June 2024

Honors: Magna Cum Laude (GPA:4.42), Ignatian Scholar (Merit Scholarship)

TECHNICAL SKILLS

Languages: C++, Python (pandas), Java, JavaScript, Bash, SQL

Embedded/FPGA: Cyclone V (Intel/Altera), Quartus Prime Lite, RTL/top-level integration, HPS-FPGA integration, AXI, memory-mapped I/O (mmap), register-level control, FSMs, timers/counters, PWM

Embedded Linux & Tools: Linux user-space peripheral control, debugging, Git, KiCad

Hardware/Test: ADALM2000 (oscilloscope/logic analyzer), NI USB-DAQ, LTspice, KiCad, breadboarding/debug, AD627 instrumentation amplifier, Arduino

Protocols: I2C, SPI

Backend/Cloud (Secondary): Node.js, Express, REST APIs, MySQL; AWS (S3, RDS, Redshift), Docker

Certifications: AWS Certified Cloud Practitioner; AWS Certified AI Practitioner

PROFESSIONAL EXPERIENCE

Khoury College of Computer Sciences, Boston, MA — Teaching Assistant — Logic & Computation

January 2026 – Current

- Held office hours to assist 25+ students with formal logic, property-based testing, and proof techniques.

Northeastern University (Knack), Boston, MA — Tutor — Differential Equations & Physics

March – December 2025

- Tutored students in Differential Equations and Physics, ensuring clarity of technical concepts.

Optimus Health Analytics, New York, NY — Data Engineer Intern

June – August 2023

- Processed and validated healthcare datasets containing ~1,000,000 records using pandas, SQL Workbench, and Excel.
- Automated ETL workflows using Bash and cron on Linux servers, reducing manual processing time by ~20%.
- Reduced ingestion failures ~30% by scripting per-load data validation and type normalization (missing fields, invalid rows) before ingestion.

TECHNICAL PROJECTS

Programmable Spider Robot | C++, Embedded Linux, FPGA Cyclone V

December 2025

- Developed gait-control logic in C++ using OOP to manage hexapod movement behaviors on a DE1-Nano.
- Diagnosed and resolved startup failures caused by timing delays in the initialization sequence, enabling consistent robot operation.

Analog ECG Signal Acquisition System | LTspice, NI USB-DAQ, MATLAB

November - December 2025

- Designed an analog front end using an AD627 amplifier with a gain of ~775.
- Implemented hardware bandpass filters (0.1–250 Hz) and digital post-processing in MATLAB (60 Hz notch, Parks-McClellan FIR); acquired live cardiac signals (~70 BPM) and evaluated filter performance against design specifications.

FPGA Motor Control Lab | Quartus Prime Lite, Cyclone V

November 2025

- Implemented PWM generation for RC Servos (50MHz clock, 20ms period) and DC motors.
- Developed a finite-state machine (FSM) for stepper-motor control with parameterized step rates.
- Built a Linux user-space controller using mmap and the AXI bus to write memory-mapped registers.

Ultrasonic Ranging System | FPGA (Cyclone V), Quartus, RTL Design

October 2025

- Designed a custom RTL driver with a dual-counter timing architecture for ultrasonic distance measurement at 20 ns precision; converted clock cycles to distance (cm) for display on triple 7-segment LEDs.
- Validated hardware performance by correlating ADALM2000 oscilloscope waveforms with digital distance readouts.

Stand AI (Oasis Program) | Node.js, Express, OpenAI API | [GitHub](#)

January – March 2025

- Built a backend API for an LLM fact-checking tool producing labels, confidence scores, and citations.
- Owned full backend development, including API architecture, endpoint design, and integration with the OpenAI API.

ACTIVITIES

- NU Wireless Club — KiCad PCB design. AI @ Northeastern — AI Applications (prompt engineering, REST API).